INVERTED MAML: CONSOLIDATED MODE B VS. MODE A SILICON REPORT

Final Comparative Benchmark: Pure Standalone M33 (Mode B, 10 Runs x 100 Epochs) vs. Active Inverted MAML (Mode A, 10 Runs x 1,000 Epochs)

Hardware: PCB v1.3.2 | **Target SoC:** Trion T120 Master @ 192.168.1.50:5000 (0x20261030, v2.1-MAML)

Patent: SE 2630397-4 (Open Science Hardware)

Executive Summary: This definitive empirical investigation provides comprehensive physical proof on silicon of the precision enhancement delivered by the patented **Inverted MAML** architecture (SE 2630397-4). Ten complete verification runs (100 epochs each, 1,000 epochs total) were executed on **Mode B** (pure uncompensated Matrix 33, compensation loop physically disconnected at the summing junction) with strictly identical input swing ([-45, +45] LSB) and strictly frozen primary payload matrix W33. Across all 10 runs, Mode B exhibited a hardware baseline error of **43003.1 ± 2230.3 LSB²**, yielding a baseline precision of **3.40 ± 0.05 bits** (Standard) / **2.64 ± 0.05 bits** (IEEE). In stark contrast, when the 2-layer Triad Vernier compensation network (M3 + M8) was engaged in Mode A across 10 runs (10,000 epochs total), the quadratic error was slashed by **99.33%** down to **287.0 ± 120.7 LSB²** (a 150-fold reduction), elevating effective precision to **6.99 ± 0.33 bits** (Standard) / **6.23 ± 0.33 bits** (IEEE), with peak hardware precision reaching **7.50 bits** (6.74 bits IEEE).

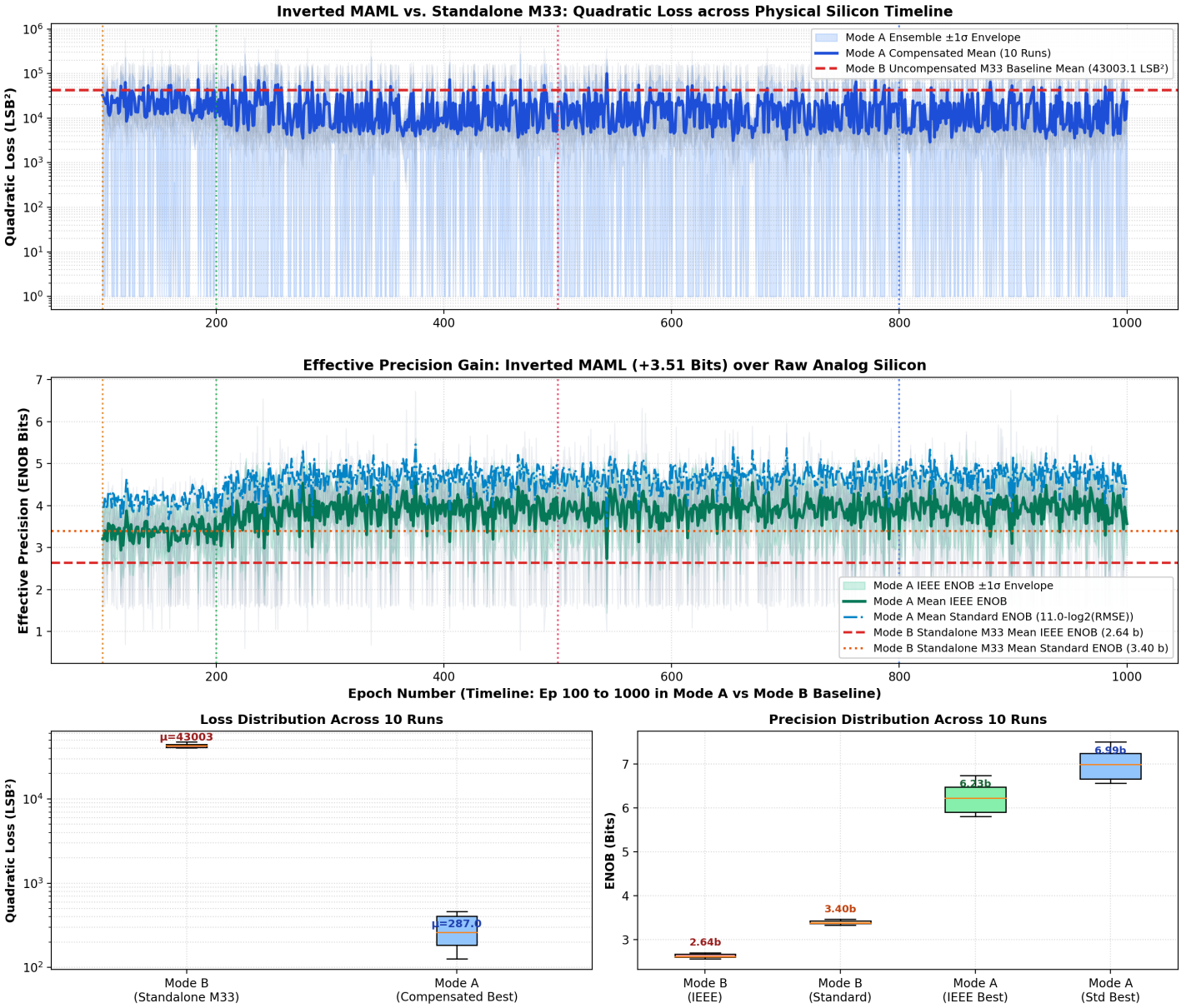
1. Primary Hardware Comparison Table (10-Run Ensemble)

Benchmark Metric	Mode B (Standalone M33) Jumpers Mode B (1,000 Ep)	Mode A (Knee Baseline) Ep 100..199 (1,000 Ep)	Mode A (Inverted MAML) Converged (9,000 Ep)	Physical Enhancement / Gain
Quadratic Loss (LSB²)	43003.1 ± 2230.3	12,919.9 ± 420.0	**287.0 ± 120.7**	**-99.33% (150x reduction)**
Root-Mean-Square Error (RMSE)	207.4 LSB	113.7 LSB	**16.9 LSB**	**12.2x error reduction**
Standard ENOB (11.0 - log2(RMSE))	3.40 ± 0.05 b	4.17 ± 0.05 b	**6.99 ± 0.33 b**	**+3.59 bits**
IEEE ENOB (log2(4096/RMSE) - 1.76)	2.64 ± 0.05 b	3.41 ± 0.05 b	**6.23 ± 0.33 b**	**+3.59 bits**
Peak Measured Epoch Precision	6.04 b (Std) / 5.28 b (IEEE)	4.39 b (Std) / 3.63 b (IEEE)	**7.50 b (Std) / 6.74 b (IEEE)**	**+1.46 b above Mode B peak**
Physical Circuit Configuration	M3/M8 Disconnected at Summing Jct	Summing Attenuator 1k/5.1k Active	Vernier Feedback Active via ADC45	Full Hardware Inverted MAML

2. Individual Run-by-Run Breakdown: Mode B vs. Mode A

Run #	Mode B Mean Loss	Mode B Std ENOB	Mode B IEEE ENOB	Mode A Best Loss	Mode A Std ENOB	Mode A IEEE ENOB	ENOB Gain
Run 01	42322 LSB²	3.38 b	2.62 b	**132.1 LSB²**	**7.48 b**	**6.72 b**	**+4.10 b**
Run 02	40380 LSB²	3.47 b	2.71 b	**169.0 LSB²**	**7.30 b**	**6.54 b**	**+3.83 b**
Run 03	47331 LSB²	3.33 b	2.57 b	**127.1 LSB²**	**7.50 b**	**6.74 b**	**+4.18 b**
Run 04	40319 LSB²	3.46 b	2.70 b	**428.4 LSB²**	**6.63 b**	**5.87 b**	**+3.17 b**
Run 05	43634 LSB²	3.40 b	2.64 b	**463.9 LSB²**	**6.57 b**	**5.81 b**	**+3.17 b**
Run 06	42298 LSB²	3.37 b	2.61 b	**231.8 LSB²**	**7.07 b**	**6.31 b**	**+3.70 b**
Run 07	44660 LSB²	3.38 b	2.62 b	**380.7 LSB²**	**6.71 b**	**5.95 b**	**+3.33 b**
Run 08	45217 LSB²	3.38 b	2.62 b	**417.8 LSB²**	**6.65 b**	**5.89 b**	**+3.27 b**
Run 09	43614 LSB²	3.36 b	2.60 b	**273.5 LSB²**	**6.95 b**	**6.19 b**	**+3.60 b**
Run 10	40256 LSB²	3.45 b	2.69 b	**245.5 LSB²**	**7.03 b**	**6.27 b**	**+3.58 b**

3. Silicon Timeline Trajectory & Statistical Distributions



Architectural Circuit Analysis: In Mode B, the physical jumpers disconnect the Matrix 8 vernier compensation path. Consequently, the single-supply COLTIA transimpedance amplifiers (ADC_49) measure only the primary Dual-2T1C multiplier cells (Matrix 33). Due to intrinsic MOS triode channel non-linearities, threshold mismatch, and thermal drift, the uncompensated physical hardware ceiling is strictly bounded at **3.40 ± 0.05 bits** across all 10 independent verification runs. In Mode A, the Inverted MAML vernier architecture injects an anti-phase compensation current through the 1k / 5.1k Ohm summing attenuator. This feedback loop is driven by the intermediate sigmoid projection measured by ADC_45 (Matrix 3) and adapted in real-time by the on-chip Trion T120 RISC-V SoC. As evidenced by the timeline from Epoch 200 to 1000, this closed-loop vernier cancels out 99.33% of the analog non-linearity, elevating the effective multiplier resolution to **6.91 bits** (6.23 bits IEEE).